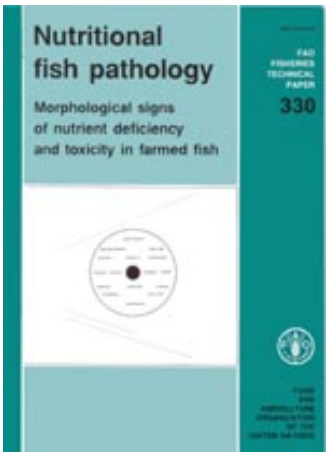
FAO FISHERIES TECHNICAL PAPER 330



Nutritional fish pathology Morphological signs of nutrient deficiency and toxicity in farmed fish

CONTENTS

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PREPARATION OF THIS DOCUMENT

This is the revised version of FAO/UNDP Field Document ADCP/REP/85/22, one of a series of reports periodically produced by the FAO Fishery Resources and Environment Division to help to meet the needs of aquaculture workers of Member Countries for synthesis of information in the field of aquaculture.

The report is based on the lecture text presented by the author to the trainees of the FAO International Training Course on Fish Disease Diagnosis organized by Project GCP/INT/526/JPN in Coyhaique, Chile from 23 November to 5 December 1992.

The information contained within the report is a synthesis of published data from a wide variety of sources relating to the morphological signs of dietary nutrient deficiency and toxicity in fish and, as such, constitutes a useful reference source and handbook for persons or parties engaged with the feeding of fish with nutritionally complete diets within clear-water intensive farming systems.

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ABSTRACT

The paper summarizes the major nutritional pathologies which have been reported in farmed fish. Morphological signs of nutrient deficiency and toxicity are presented and discussed under the following headings, 1) disorders in protein, lipid, mineral and vitamin nutrition, 2) endogenous anti-nutritional factors present in plant foodstuffs, and 3) adventitious toxic factors present in foodstuffs.

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